The LUR quasihyperbolic-ball question in arXiv:1407.2403

Explicitly answered by Theorem 3.2 of arXiv:1708.02240

Automated Mathematics Run fa_banach_001, lane 2

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Status. literature_already_answered. This is an exact, author-acknowledged later answer, not a new solution from the run.

Original question

Klén, Rasila, and Talponen ask on source PDF page 15, at the end of Section 4 of *On smoothness of quasihyperbolic balls* [1], whether centered quasihyperbolic balls in symmetric convex domains of reflexive LUR Banach spaces induce LUR norms through their Minkowski functionals. In the notation of the paper, for $B = B_k(0, r)$ this functional is

$$M_B(x) = \inf\{\lambda > 0 : x \in \lambda B\}.$$

Exact later answer

Rasila, Talponen, and Zhang state in their abstract that they answer a question posed by the first two authors with R. Klén [2]. Immediately before Theorem 3.2 on supporting PDF page 5, they identify the LUR Minkowski question from [1] and say it is settled affirmatively in the reflexive case. Theorem 3.2 proves the stronger sequential statement: if X is reflexive and LUR, $\Omega \subset X$ is convex, and

$$k(x_0, y) = k(x_0, y_n), \qquad k\left(x_0, \frac{y + y_n}{2}\right) \longrightarrow k(x_0, y),$$

then $y_n \to y$ in norm. Its final sentence gives the requested conclusion verbatim in mathematical content: if Ω is symmetric, the Minkowski functional of $B_k(0, r)$ is an equivalent LUR norm for every $r > 0$.

Thus the question extracted from arXiv:1407.2403 has a complete positive literature answer in Theorem 3.2 of arXiv:1708.02240. The overlap is explicit, not an agent-inferred reformulation.

Search bounds and scope

The run's lightweight indexes had no packet for this exact question. A bounded search through 19 July 2026 used arXiv:1407.2403, the exact question phrase, "quasihyperbolic balls LUR Minkowski functional", and "reflexive LUR quasihyperbolic". It found arXiv:1708.02240 and no need for a new proof. The supporting theorem exactly covers the original reflexive-LUR question; no part of that stated question remains open in this packet.

References

[1] R. Klén, A. Rasila, and J. Talponen, *On smoothness of quasihyperbolic balls*, Annales Academiae Scientiarum Fennicae Mathematica 42 (2017), 439–452; arXiv:1407.2403.

[2] A. Rasila, J. Talponen, and X. Zhang, *Observations on quasihyperbolic geometry modeled on Banach spaces*, arXiv:1708.02240 (2017), Theorem 3.2.